

Community Pharmacist–Provided Extended Diabetes Care

William R Doucette, Matthew J Witry, Karen B Farris, and Randal P McDonough

Diabetes is a common and costly disease that affects more than 7.8% of the US population, with an estimated annual cost exceeding \$174 billion.¹ Diabetes is a difficult disease to manage because of the numerous self-care behaviors involved in controlling blood glucose levels and other disease aspects. For patients with diabetes, failure to control blood glucose levels is associated with microvascular complications such as nephropathy, neuropathy, and retinopathy.^{2,3} Controlling cardiovascular risk factors is also important, because patients with diabetes are at an increased risk of myocardial infarction and stroke.³ Diabetes is a progressive disease that, once diagnosed, requires lifelong management. Patients diagnosed with diabetes often are prescribed new medications and educated on important lifestyle changes. Supplementary diabetes management beyond this initial period is referred to herein as extended diabetes care.

Managing diabetes can be difficult, especially for a single provider. Often, physician visits are short and focus on an acute complaint.⁴ As a result, physicians may not have enough time to assess drug therapy or provide diabetes education, especially considering the significant lifestyle changes recommended for patients with diabetes.^{5,6} Including additional providers such as pharmacists in the care of patients with diabetes has been shown to be beneficial. For example, several studies have shown that

BACKGROUND: Pharmacists in various settings have been effective in initiating diabetes treatment. Patients with diabetes require ongoing disease management, and community pharmacists are in a strategic position to provide such extended care. Little is known, however, about the effects of community pharmacist–provided interventions beyond the initial treatment period.

OBJECTIVE: To evaluate the effect of community pharmacist–provided extended diabetes care service on primary clinical outcomes, including hemoglobin A_{1c} (A1C), low-density lipoprotein cholesterol (LDL-C), and blood pressure, and on patients' reported self-care activities.

METHODS: A randomized controlled trial was conducted in patients with diabetes. Participants had already completed at least 2 diabetes education sessions at a local diabetes education center. Nine specially trained pharmacists administered interventions during up to 4 quarterly visits per patient. Interventions included discussing medications, clinical goals, and self-care activities with patients and recommending medication changes to physicians when appropriate. The main outcome measures were 12-month changes in A1C, LDL-C, blood pressure, and self-report of self-care activities.

RESULTS: Seventy-eight patients participated in the study (36 intervention, 42 control); 66 were included in the final analysis (31 intervention, 35 control). Compared with changes in the control group, patients who received interventions significantly increased the number of days per week that they engaged in a set of diet and diabetes self-care activities (1.25 and 0.73 more days/wk, respectively). The mean 12-month changes for A1C, LDL-C, and blood pressure were not significantly different between the 2 study groups.

CONCLUSIONS: Although pharmacist-provided interventions did not demonstrate statistically significant improvements in clinical outcomes over the study period, study results did show that pharmacists were effective at increasing the number of days that patients spent engaging in healthy diet and diabetes self-care activities. Addressing lifestyle and self-care behaviors can be a beneficial component of a pharmacist-provided extended diabetes care service.

KEY WORDS: diabetes, disease management, medication therapy management, pharmacy.

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pharmacist interventions can decrease hemoglobin A_{1c} (A1C) levels by 1% or more.⁷ The American Diabetes Association has affirmed this and endorses multidisciplinary diabetes care.⁸ Some insurers recognize the need for additional instruction about diabetes self-care and pay for spe-

Author information provided at the end of the text.

cialized education sessions. However, the number of visits that a beneficiary may receive typically is limited.

Patients may benefit from additional diabetes education for several reasons. First, diet and exercise interventions can help decrease the A1C level and obtain other health benefits.^{9,10} Second, at the time of diagnosis, many patients are not ready to make certain lifestyle changes.^{11,12} For example, it may be years before some patients are ready to consider changing their diet or exercising regularly. Further complicating the process, patients are asked to make multiple changes simultaneously and maintain these new behaviors indefinitely. Providing consistent education and reinforcement has been shown to help patients sustain lifestyle changes.¹² Lastly, new issues such as insulin therapy may emerge as a patient's condition progresses. Multidisciplinary extended diabetes care allows patients to receive customized diabetes education and motivation as their outlook, readiness, and condition change.

An extended diabetes care service also may be valuable from a clinical perspective. Despite strong evidence that controlling clinical markers such as blood pressure, low-density lipoprotein cholesterol (LDL-C), and A1C decreases morbidity and mortality, the US healthcare system struggles to help patients meet their goals.¹ Healthcare professionals offering extended diabetes care can collaborate with physicians to optimize patient medication regimens, with a shared goal of improving outcomes and decreasing morbidity.

For these reasons, patients with diabetes may benefit from receiving extended diabetes care consisting of education and medication management components. Community pharmacists are in a strategic position to provide extended diabetes care for several reasons. Pharmacists may see a patient with diabetes 5 times more often than does his or her primary care physician.¹³ Pharmacists have significant training in the pharmacotherapy of diabetes and cardiovascular disease and can make valuable drug therapy recommendations to physicians. Pharmacists also are skilled at identifying adherence problems and addressing adverse drug effects.¹⁴ Lastly, many pharmacists are trained to educate and motivate patients on diet, exercise, and diabetes self-care activities.

One meta-analysis showed that pharmacist interventions can significantly decrease A1C levels across a variety of settings, including clinics and community pharmacies.¹⁵ Likewise, several trials have demonstrated that pharmacist interventions can decrease lipid levels and blood pressure.¹⁶⁻²¹ Many of the subjects in these studies were newly diagnosed or patients with uncontrolled type 2 diabetes, and the interventions consisted primarily of starting drug therapy and providing initial diabetes education.

The effects of community pharmacist-provided interventions on A1C have been mixed.²²⁻²⁴ A randomized con-

trolled trial of a pharmacotherapy follow-up program in Spain for patients with diabetes showed a statistically significant 1.2% improvement in A1C in the intervention group compared with the control group.²² Participants receiving this intervention also demonstrated improvements in diabetes knowledge and blood pressure. A prospective cohort study using community pharmacists in Australia showed an improvement in A1C of 0.47% in the intervention group compared with the control group, although the change was not statistically significant ($p = 0.07$).²³ In a randomized controlled trial of a Canadian pharmacist-provided diabetes care program, intervention patients had a nonsignificant 0.2% improvement in A1C compared with control ($p = 0.57$).²⁴

In the Asheville Project, community pharmacists provided extended diabetes care consisting of education and medication management for patients beyond their initial diagnosis and treatment period.²⁵ The Asheville Project showed significant clinical improvements, including decreases in A1C, and cost savings, following community pharmacist interventions; however, it did not use a randomized controlled design, nor did it control for outside sources of diabetes education. There is a need to study the effects of a community pharmacist-provided extended diabetes care model using a randomized controlled design.

Our goal was to evaluate the effect of a community pharmacist-provided extended diabetes care service on primary clinical outcomes, including A1C, LDL-C, and blood pressure, and on patients' reported self-care activities.

Methods

This study used a randomized controlled design to assess the effect of a community pharmacist-provided diabetes management intervention on changes in patient-reported behaviors and clinical markers. Researchers randomized adults with type 2 diabetes who had completed at least 2 diabetes education sessions at a local diabetes education center within the past 2 years to receive either usual care or extended diabetes care from a pharmacist at one of the 7 study pharmacies. To be eligible for the study, patients were required to have an A1C of 7.0% or greater as determined by the most recent laboratory test results. Exclusion criteria included dialysis, hepatic disorder, stage IV heart failure, severe ischemic/hemorrhagic stroke, legal blindness, diabetes-related amputation, gestational diabetes only, and dementia. Focusing on A1C, a sample size of 50 patients per study group was calculated, using a medium effect size of 0.5, α of 0.05, and power of 0.80. The effect size of 0.5 for a pharmacist-based diabetes management service is in the middle of the range of effect sizes calculated from studies of such services reported in a review by Wubben and Vivian.⁷

Participating pharmacists received training in diabetes management and study protocol. The diabetes management training included both self-study and live programs. The self-study component was a 15-hour certificate program in diabetes management called "Pharmaceutical Care for Patients with Diabetes." The live training covered the pathophysiology of diabetes, therapeutics of diabetes medications, diabetes self-care, and discussions of mock cases. In addition, pharmacists were required to demonstrate competency in measuring blood pressure, using a blood glucose meter, filling an insulin syringe, and administering an insulin injection. The goal of the training was to ensure that all participating pharmacists had current knowledge and competency in managing diabetes.

The pharmacist's role in the interventions involved a 5-step process of care: gathering information from patients and other sources, evaluating the information, formulating a plan, implementing the plan, monitoring the plan, and following up with the patient and physician to ensure optimal outcomes. Pharmacists were instructed to assess clinical parameters such as A1C, LDL-C, and blood pressure, and then to use these values to educate patients and recommend drug therapy changes. Pharmacists also were to assess self-care activities and make recommendations when appropriate. Overall, pharmacists had flexibility to customize interventions to the needs of the patients.

Patients in the extended-care group received up to 4 (quarterly) visits with 1 of 9 specially trained community pharmacists at their community pharmacy practice site. During the first visit, the study protocol recommended that pharmacists take a patient history, create a medication list, assess clinical markers, review medications and self-care behaviors, and identify drug therapy problems. Subsequent visits were intended to allow pharmacists to follow-up on previous problems, identify new problems, reassess clinical parameters such as blood glucose and blood pressure, and discuss self-care activities. After the visits, pharmacists faxed a one-page progress note to patients' physicians describing the content of the visit. Progress notes followed a subjective-objective-assessment-plan (SOAP) format. Pharmacists could use SOAP notes to recommend drug therapy changes for the physician to consider. Patients in the control group received usual diabetes care from their primary care provider. Study participants did not receive additional diabetes education sessions from the participating diabetes education center during the study period.

Patients presented to the diabetes center at baseline and 12 months for data collection. Data included blood pressure readings taken by the research nurse, as well as A1C and lipid profiles from the clinical laboratory or physician's office. Patients also completed questionnaires at the baseline and 12-month visits. The self-care items, taken from a 2000 study by Toobert et al.,²⁶ assessed how many days out of the past 7 that patients engaged in each of 10

self-care activities related to diabetes management (Appendix I). Three items assessed good dietary habits, 2 related to exercise, and 5 activities were specific to managing diabetes. To analyze the questionnaire results, researchers calculated the mean responses for each category: diet, exercise, and diabetes. This created 3 activity sets, each with a possible value of 0–7. These values corresponded to the mean number of days that patients reported engaging in the activities comprising a given category. Higher values corresponded to reporting more frequent participation in healthy activities.

Post hoc, progress notes were manually reviewed from a convenience sample of 3 of the 7 pharmacies. Researchers examined progress notes to describe different education points covered, note the frequency of medication recommendations made to physicians, and record how frequently physicians responded.

Researchers used SPSS 12.0 (SPSS Inc., Chicago, IL) to analyze the impact of the intervention by comparing 12-month changes between the intervention and control groups. Clinical measures included A1C and LDL-C, as well as systolic and diastolic blood pressure. Changes in questionnaire responses for diet, exercise, and diabetes self-care activities were also compared. Independent sample *t*-tests or χ^2 tests compared the 2 groups at baseline. Paired *t*-tests were used to compare baseline and 12-month values within the study groups. Independent 2-sample *t*-tests were used to test the difference between the changes in the intervention group versus the control group. This study was approved by the Human Subjects Office at the University of Iowa.

Results

There were 160 eligible patients at the diabetes center; 6 could not be contacted. Of the 154 patients asked to participate in the study, 78 consented and were randomized to either the intervention group (*n* = 36) or the control group (*n* = 42) and were presented to the diabetes center for baseline data collection (Table 1). Of the subjects who started the study, 4 died, 6 were unable or unwilling to do the final data collection visit and/or laboratory tests, and 2 subjects randomized to the intervention group failed to meet with a study pharmacist for the intervention. As a result, 66 subjects (31 in the intervention group and 35 in the control group) presented to the diabetes center for final data collection and were used in the comparative analysis (Table 2). For the 31 intervention subjects completing the final data collection, 87.1% of the possible pharmacist visits occurred: 24 patients received all 4 visits, 2 patients received 3 visits, 1 patient received 2 visits, and 4 patients received 1 visit.

There were no significant differences between the demographics of the control and intervention groups at baseline (Table 1). The overall mean baseline A1C was 7.95%, which is above the target of 7.0% recommended by the

American Diabetes Association.⁸ Similarly, the baseline overall mean LDL-C level of 102.5 mg/dL exceeded the target of 100 mg/dL established by the Adult Treatment Panel III of the National Cholesterol Education Program.²⁷ Conversely, baseline mean blood pressures, 118.2 mm Hg systolic and 66.7 mm Hg diastolic, were lower than the target of 130/80 mm Hg established by the Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure.²⁸

The mean 12-month changes for the 3 clinical markers (A1C, LDL-C, blood pressure) were not significantly different between the study groups (Table 2). The intervention group trended toward improvements in A1C ($p = 0.27$) and LDL-C ($p = 0.32$). Of note, both the control and intervention groups showed increases in systolic blood pressure and statistically significant decreases in LDL-C. The mean increase in systolic blood pressure was 4.5 mm Hg for the control group ($p = 0.254$) and 7.1 mm Hg for the treatment group ($p = 0.002$). The mean decrease in LDL-C was 12.0 mg/dL for the control group ($p = 0.013$) and 19.6 mg/dL for the intervention group ($p = 0.003$).

Subjects in the intervention group significantly increased their reported engagement in healthy self-care ac-

tivities. Compared with subjects in the control group, those in the intervention group significantly increased the number of days per week that they engaged in diet and diabetes self-care activities (Table 2 and Appendix I). The set of exercise self-care items was not significantly affected by the intervention.

The post hoc manual chart review included 20 patient charts (57% of the intervention group) and 58 visit notes (50% of the interventions) from 3 of the 7 study pharmacies. Pharmacists commonly educated patients about medication adherence, diet, exercise, and blood glucose monitoring. Other topics included immunizations, laboratory test results and goals, and issues not directly related to diabetes. Of the 58 SOAP notes sent to physicians, 14 (24%) included medication recommendations or dose changes; of these, 4 recommendations were accepted by physicians.

Discussion

The intervention in this study produced mixed results. Patients who received pharmacist-provided extended diabetes care reported engaging in more healthy behaviors, specifically activities related to diet and diabetes self-care.

Table 1. Patient Baseline Characteristics

Characteristic	Intervention Group, n (%) (n = 36)	Control Group, n (%) (n = 42)	Total, n (%) (N = 78)	p Value
Age, y (mean $\pm$ SD)	58.7 $\pm$ 13.3	61.2 $\pm$ 10.9	60.0 $\pm$ 12.0	0.373 ^a
Female	21 (61.8)	22 (53.7)	43 (57.3)	0.598 ^b
White	34 (100)	41 (100)	75 (100)	1.00 ^b
High school graduate or less	22 (64.7)	25 (64.1)	47 (64.4)	0.957 ^b
College or technical degree	12 (35.3)	14 (35.9)	26 (35.6)	0.957 ^b
Household income, \$				
<25,000	15 (45.5)	9 (24.3)	24 (34.2)	0.157 ^b
25,000–54,999	12 (36.4)	18 (48.6)	30 (42.9)	0.157 ^b
$\geq$ 55,000	6 (18.2)	10 (27.0)	16 (22.9)	0.157 ^b
Married	20 (58.8)	29 (70.7)	49 (65.3)	0.219 ^b
Smoke tobacco	4 (13.3)	3 (7.9)	7 (10.3)	0.541 ^b
Drink alcohol	17 (50.0)	20 (48.7)	37 (49.3)	0.972 ^b
BMI, kg/m ² (mean $\pm$ SD)	35.5 $\pm$ 8.22	34.5 $\pm$ 6.83	34.9 $\pm$ 7.44	0.568 ^a
A1C, % (mean $\pm$ SD)	7.99 $\pm$ 1.45	7.91 $\pm$ 1.91	7.95 $\pm$ 1.70	0.849 ^a
LDL-C, mg/dL (mean $\pm$ SD)	100.7 $\pm$ 28.04	105.1 $\pm$ 36.55	103.1 $\pm$ 32.83	0.563 ^a
Systolic BP, mm Hg (mean $\pm$ SD)	118.2 $\pm$ 11.67	119.8 $\pm$ 17.62	119.1 $\pm$ 15.14	0.665 ^a
Diastolic BP, mm Hg (mean $\pm$ SD)	66.6 $\pm$ 8.90	67.1 $\pm$ 8.32	66.9 $\pm$ 8.53	0.778 ^a
Diet self-care activities, days (mean $\pm$ SD) ^c	3.86 $\pm$ 1.41	4.38 $\pm$ 1.34	4.14 $\pm$ 1.39	0.107 ^a
Diabetes self-care activities, days (mean $\pm$ SD) ^d	4.64 $\pm$ 1.76	5.06 $\pm$ 1.33	4.87 $\pm$ 1.54	0.240 ^a
Exercise self-care activities, days (mean $\pm$ SD) ^e	2.68 $\pm$ 2.13	3.65 $\pm$ 2.48	3.21 $\pm$ 2.36	0.077 ^a

A1C = hemoglobin A_{1c}; BMI = body mass index; BP = blood pressure; LDL-C = low-density lipoprotein cholesterol.

^aIndependent t test of percent of intervention group vs percent of control group.

^b χ^2 test of frequency of intervention group vs control group.

^cMean days per week engaging in 3 diet self-care activities.

^dMean days per week engaging in 5 diabetes self-care activities.

^eMean days per week engaging in 2 exercise self-care activities.

However, these changes did not translate into significant improvements in A1C, LDL-C, or blood pressure when compared with control patients. Studies of lifestyle modifications have demonstrated that adopting healthy diet and exercise habits can improve clinical markers such as blood pressure, A1C, and LDL-C.^{9,10,29,30} As a result, lifestyle modifications are the initial recommendation of the major clinical guidelines for diabetes, hypertension, and cholesterol and should complement drug therapy.^{8,27,28}

At the conclusion of the trial, the mean number of days per week that patients reported engaging in the 3 diet activities increased by 1.25 days versus control. There are several explanations for why this change was not reflected in clinical markers. One possibility is that patients' diet habits improved gradually over the course of the study, or possibly not until the end. While diet changes made toward the end of the study would have been reported on the final survey, there may not have been sufficient time for these behaviors to fully induce physiological changes that would be reflected in the clinical markers. Secondly, the impact of the changes may have been too small to precipitate an improvement in clinical markers, given the small size of the study. Regardless, the improvement in reported diet self-care activities suggests that education on diet should continue to be a component of pharmacist-provided diabetes care.

Patients in the intervention group also reported improving their participation in a group of 5 diabetes management activities, whereas the control group reported a worsening in participation. At the end of the study, the intervention group reported increasing the mean number of days per week per activity for the set of 5 diabetes management activities by 0.73 days versus the control group. These activities included 2 items on blood sugar monitoring, 2 items

on foot care, and 1 item on medication adherence. Of these items, only medication adherence is likely to have a direct effect on the aforementioned clinical markers. Therefore, improvements in this scale may not be reflected by clinical markers.

Important clinical markers did not show statistically significant improvements. The stasis of A1C levels in our study was disappointing, especially since a meta-analysis showed that this marker was sensitive to pharmacist interventions.¹⁵ This may be partly due to differences in patient treatment histories. Patients in our study already had begun treatment for diabetes and had participated in 2 diabetes education classes. As a result, they had a lower mean baseline A1C value (7.95%) compared with the A1C reported in the meta-analysis (8.72%). It may become more difficult to lower A1C levels as patients get closer to goal because of the increased burden on patients to further improve diet and exercise or take additional medications. Alternatively, further intensifying glucose management may have been inappropriate for some patients due to the risk of hypoglycemia. It is unknown how frequently this was the case.

Another important risk factor for patients with diabetes is high blood pressure. In our study, mean blood pressure readings increased for both the control and intervention groups, though there was no statistical difference between groups. Even with these increases, mean systolic and diastolic blood pressure readings for both groups remained below goal.²⁸ The increase seen in the treatment group is not necessarily a shortcoming of the intervention, because there would have been no reason for a study pharmacist to intervene when a subject's blood pressure is already below goal. Pharmacists can play a key role in the management of hypertension by consistently monitoring blood pres-

Table 2. Comparison of Clinical Variables

Variable	Change from Baseline		Absolute Difference (intervention – control)	p Value ^a
	Intervention Group, Mean ± SD (range) (n = 31)	Control Group, Mean ± SD (range) (n = 35)		
A1C (%)	-0.27 ± 1.11 (-2.8 to 2.3)	0.12 ± 1.73 (-6.10 to 4.30)	-0.39	0.272
LDL-C (mg/dL)	-19.6 (-106 to 32)	-12.0 (-83 to 29)	-7.6	0.320
Systolic BP (mm Hg)	7.1 ± 10.38 (-10 to 28)	4.5 ± 15.19 (-17 to 62)	2.6	0.367
Diastolic BP (mm Hg)	1.2 ± 7.79 (-15 to 19)	0.3 ± 8.45 (-19 to 26)	0.9	0.705
Diet self-care activities (days) ^c	0.70 ± 1.39 (-2.0 to 4.0)	-0.55 ± 1.36 (-3.67 to 2.33)	1.25	0.001 ^b
Diabetes self-care activities (days) ^d	0.79 ± 1.48 (-1.8 to 5.6)	0.06 ± 0.93 (-2.0 to 1.8)	0.73	0.027 ^b
Exercise self-care activities (days) ^e	0.58 ± 2.06 (-4.0 to 5.0)	0.10 ± 1.65 (-3.5 to 5.0)	0.48	0.312

A1C = hemoglobin A_{1c}; BP = blood pressure; LDL-C = low-density lipoprotein cholesterol.
^aIndependent 2-sample *t* test used to compare 12-month changes between intervention and control groups; equal variances assumed.
^bp Value significant at <0.05.
^cMean days per week engaging in 3 separate diet self-care activities.
^dMean days per week engaging in 5 separate diabetes self-care activities.
^eMean days per week engaging in 2 separate exercise self-care activities.

tures and making recommendations to physicians when patients are above goal.

Both the intervention and control groups showed substantial decreases in LDL-C; however, there was no statistically significant difference between groups. These decreases are likely due to lipid-lowering drug therapy. Lipid management is an important component of diabetes care and hydroxymethylglutaryl coenzyme A reductase inhibitors are commonly initiated with the diagnosis of diabetes and titrated soon after. In addition, when this study was conducted, new evidence supporting stricter LDL-C goals for patients with diabetes was disseminating through the medical community.³¹ Thus, the subjects' decreases in LDL-C may be due to changes in physician prescribing patterns.

The SOAP note sample may help to explain why the intervention group reported engaging in more healthy activities but did not show improvements in clinical markers versus the control group. The majority of interventions focused on patient education and motivation, not medication changes. Pharmacists frequently discussed diet, exercise, medication adherence, blood glucose monitoring, and other important aspects of diabetes self-care with patients. Unfortunately, there may not have been a sufficient number of visits over a long enough study period to significantly affect clinical markers, especially considering that health-related behaviors are often difficult for patients with diabetes to change and such change may occur slowly, one behavior at a time. Along similar lines, interventions with a more robust focus on optimizing patient medication regimens may show greater improvement in clinical markers.

There is room for improvement on this model. Pharmacists, as medication experts, should ensure that their services contain a strong medication management component. Pharmacist-provided diabetes management services appear to have the greatest clinical impact when focused on medication optimization, especially when the pharmacists are given prescriptive authority as part of a collaborative practice agreement.⁷ Pharmacists in our study communicated low numbers of drug therapy adjustments to physicians; this may be related to the fact that some of the pharmacists in the study had little prior experience collaborating with physicians. With experience, pharmacists may become more effective and confident in recommending medication changes to physicians.

This study had a relatively small sample size due in part to the withdrawal of a second clinic around the start of patient recruitment; this constrained the pool of eligible participants. While about 50% of eligible patients were recruited, the overall pool was not large enough to reach the targeted sample size. The decreased number of participants limited the power of the study to detect small, but potentially meaningful, differences in clinical markers. Also, there were several dropouts, and 6 intervention patients received only 1 of 4 study visits. This may have blunted the

overall impact of the intervention. Secondly, the study protocol was flexible, in that pharmacists were able to decide when and how often to propose medication changes. Such a protocol allowed pharmacists to focus on patient education instead of proposing medication changes. This may have resulted in pharmacists providing conservative interventions that limited the effects of this extended diabetes care program. Lastly, the study population may not be representative of the general population of patients with diabetes. Since participants were recruited from patients of a diabetes education center, there may be self-selection for patients with greater motivation and self-efficacy.

This study showed that community pharmacists can effectively help patients to increase their participation in healthy diabetes self-care activities. Further development of the pharmacist-provided extended diabetes care model is needed. A follow-up study with more participants, a longer study period, and greater focus on medication management would help to further evaluate the benefits of pharmacist-provided extended diabetes care.

William R Doucette PhD, Professor, Division of Clinical and Administrative Pharmacy, College of Pharmacy, University of Iowa, Iowa City, IA

Matthew J Witry PharmD, Graduate Fellow, Division of Clinical and Administrative Pharmacy, College of Pharmacy, University of Iowa

Karen B Farris BSP Pharm PhD MPA, Professor, Division of Clinical and Administrative Pharmacy, College of Pharmacy, University of Iowa

Randal P McDonough MS PharmD CGP BCPS, Clinical Pharmacist, Towncrest Pharmacy, Iowa City

Reprints: Dr. Doucette, Division of Clinical and Administrative Pharmacy, College of Pharmacy, University of Iowa, S518 PHAR, Iowa City, IA 52242, fax 319/353-5646, william-doucette@uiowa.edu

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Appendix I. Diabetes Self-Care Survey Questions

INSTRUCTIONS: The questions below ask you about your diabetes self-care activities during the past 7 days. If you were sick during the past 7 days, please think back to the last 7 days that you were not sick.

For each activity circle the number of days in the past week in which you performed the activity.

On how many of the last SEVEN DAYS...

Activity	Number of days per week						
Have you followed a healthy eating plan? ^a	1	2	3	4	5	6	7
Did you eat 5 or more servings of fruits and vegetables? ^a	1	2	3	4	5	6	7
Did you eat high-fat foods such as red meat or full-fat dairy products? ^{a,b}	1	2	3	4	5	6	7
Did you participate in at least 30 minutes of physical activity? ^c	1	2	3	4	5	6	7
Did you participate in a specific exercise session (such as swimming, walking, biking) other than what you do around the house or as part of your work? ^c	1	2	3	4	5	6	7
Did you test your blood sugar? ^d	1	2	3	4	5	6	7
Did you test your blood sugar the number of times recommended by your healthcare provider? ^d	1	2	3	4	5	6	7
Did you check your feet? ^d	1	2	3	4	5	6	7
Did you inspect the insides of your shoes? ^d	1	2	3	4	5	6	7
Did you take your recommended diabetes medication? ^d	1	2	3	4	5	6	7

^aIncluded in the diet activities mean.

^bResponses to this item were subtracted from 7 (reverse coded).

^cIncluded in the exercise activities mean.

^dIncluded in the diabetes care activities mean.

Atención en Diabetes Ofrecida por el Farmacéutico de Comunidad

WR Doucette, MJ Witry, KB Farris, y RP McDonough

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EXTRACTO

TRASFONDO: Farmacéuticos en diversos escenarios han sido exitosos en la iniciación de tratamiento para la diabetes. Los pacientes con diabetes también necesitan de manejo continuo de su condición y los farmacéuticos de comunidad están en una posición estratégica para ofrecer ese tipo de atención a largo plazo. Poco se conoce, sin embargo, acerca de los efectos de las intervenciones que realiza el farmacéutico de comunidad posterior al período de tratamiento inicial. Este estudio evalúa los resultados de un servicio de atención a pacientes diabéticos ofrecido por farmacéuticos de comunidad que trasciende el período inicial de tratamiento.

OBJETIVOS: Evaluar los efectos de un servicio extendido a pacientes diabéticos ofrecido por farmacéuticos de comunidad en resultados clínicos primarios que incluyen HbA_{1c}, colesterol LDL, y presión sanguínea, y las actividades de auto-atención reportadas por los pacientes.

MÉTODOS: Estudio aleatorio con control de pacientes con diabetes. Los participantes habían completado por lo menos 2 sesiones de educación en diabetes en un centro de educación en diabetes local. Nueve farmacéuticos especialmente adiestrados tuvieron a su cargo realizar las intervenciones durante 4 visitas trimestrales por paciente. Las intervenciones incluyeron discutir el uso de medicamentos, las metas clínicas, y las actividades de auto-cuidado con los pacientes y recomendar cambios en la terapia con medicamentos a los médicos de los pacientes cuando fuera necesario. Las medidas de resultados principales fueron los cambios a 12 meses en hemoglobina glucosilada, colesterol LDL, presión sanguínea, y reportes de actividades de auto-atención.

RESULTADOS: Setenta y ocho pacientes participaron en el estudio, 36 en el grupo de intervención y 42 en el grupo control. Sesenta y seis pacientes se incluyeron en el análisis final, 31 en el grupo de intervención, y 35 en el grupo control. Comparado al grupo control, los pacientes que recibieron las intervenciones significativamente aumentaron el número de días a la semana en que se ocuparon en actividades relacionadas a la dieta y la auto-atención (1.25 y 0.73 días a la semana, respectivamente). En promedio, los cambios a 12 meses en HbA_{1c}, colesterol LDL, y presión sanguínea no fueron significativamente diferentes entre ambos grupos.

CONCLUSIONES: Aunque la intervención del farmacéutico en este estudio no resultó en mejoría estadísticamente significativa en los resultados clínicos (HbA_{1c}, colesterol LDL, y presión sanguínea) en el período de 12 meses, sí pudo demostrar que los farmacéuticos pudieron aumentar el número de días en que los pacientes se ocuparon de realizar actividades saludables en su dieta y auto-atención. Abordar conductas de estilos de vida saludables y auto-atención puede ser un componente beneficioso de un servicio extendido de atención a pacientes diabéticos ofrecido por farmacéuticos de comunidad.

Traducido por Homero A Monsanto

Pharmacien Communautaire et Prestation de Soins aux Patients Diabétiques

WR Doucette, MJ Witry, KB Farris, et RP McDonough

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RÉSUMÉ

OBJECTIF: Évaluer l'impact de pharmaciens communautaires offrant des services aux patients souffrant de diabète sur les résultats thérapeutiques primaires incluant l'hémoglobine glyquée, le cholestérol LDL, et la tension artérielle; et sur les activités d'autotraitement.

MÉTHODES: Une étude randomisée a été effectuée pour des patients souffrant de diabète. Les participants avaient déjà complété au moins 2 sessions de formation au niveau du diabète dans des centres ambulatoires en diabète. Neuf pharmaciens, ayant reçu une formation spécialisée, ont effectué les interventions ciblées. Les interventions comprenaient une discussion sur les médicaments, les objectifs thérapeutiques, les activités d'autotraitement et recommandaient lorsque nécessaire des modifications aux médecins dans les médicaments. Les objectifs primaires comprenaient un changement pendant une période d'un an de l'hémoglobine A_{1c}, du cholestérol LDL, de la tension artérielle, et des activités d'autotraitement.

RÉSULTATS: Un nombre de 78 patients ont participé à l'étude (36 dans le groupe intervention et 42 dans le groupe contrôle). Dans l'évaluation finale, 66 participants ont été inclus dont 31 dans le groupe d'intervention et 35 dans le groupe contrôle. Lorsque comparé aux changements dans le groupe contrôle, le nombre de journées par semaine auquel les participants ont adhéré à une diète et aux activités d'autotraitement ont augmenté de façon significative pour le groupe avec interventions. Les changements de l'hémoglobine glyquée, du cholestérol LDL et de la tension artérielle n'étaient pas cliniquement significatifs dans les 2 groupes.

CONCLUSIONS: Même si les interventions des pharmaciens n'ont pas démontré une amélioration cliniquement significative dans les résultats thérapeutiques pour l'hémoglobine glyquée, le cholestérol LDL, et la tension artérielle sur une période de 12 mois, cette étude a démontré que les pharmaciens ont aidé à améliorer le nombre de jours durant lequel les patients ont adhéré à une diète saine et aux activités d'autotraitement pour le diabète. Le changement dans les habitudes de vie et les comportements d'autotraitement peuvent être des composantes importantes à cibler pour les pharmaciens.

Traduit par Louise Mallet